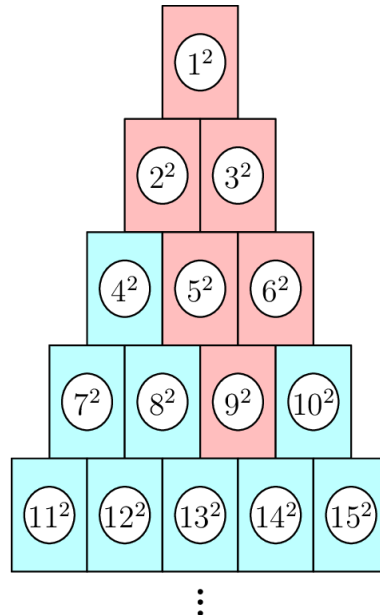


G. Hits Different

time limit per test: 2.5 seconds
memory limit per test: 256 megabytes

In a carnival game, there is a huge pyramid of cans with 2023 rows, numbered in a regular pattern as shown.



If can 9^2 is hit initially, then all cans colored red in the picture above would fall.

You throw a ball at the pyramid, and it hits a single can with number n^2 . This causes all cans that are stacked on top of this can to fall (that is, can n^2 falls, then the cans directly above n^2 fall, then the cans directly above those cans, and so on). For example, the picture above shows the cans that would fall if can 9^2 is hit.

What is the **sum** of the numbers on all cans that fall? Recall that $n^2 = n \times n$.

Input

The first line contains an integer t ($1 \leq t \leq 1000$) — the number of test cases.

The only line of each test case contains a single integer n ($1 \leq n \leq 10^6$) — it means that the can you hit has label n^2 .

Output

For each test case, output a single integer — the sum of the numbers on all cans that fall.

Please note, that the answer for some test cases won't fit into 32-bit integer type, so you should use at least 64-bit integer type in your programming language (like `long long` for C++). For all valid inputs, the answer will always fit into 64-bit integer type.

Example

input	output
10	156
9	1
1	5
2	10
3	21
4	39
5	46
6	146
10	63145186
1434	58116199242129511
1000000	

Note

The first test case is pictured in the statement. The sum of the numbers that fall is

$$1^2 + 2^2 + 3^2 + 5^2 + 6^2 + 9^2 = 1 + 4 + 9 + 25 + 36 + 81 = 156.$$

In the second test case, only the can labeled 1^2 falls, so the answer is $1^2 = 1$.

In the third test case, the cans labeled 1^2 and 2^2 fall, so the answer is $1^2 + 2^2 = 1 + 4 = 5$.

In the fourth test case, the cans labeled 1^2 and 3^2 fall, so the answer is $1^2 + 3^2 = 1 + 9 = 10$.

In the fifth test case, the cans labeled 1^2 , 2^2 , and 4^2 fall, so the answer is

$$1^2 + 2^2 + 4^2 = 1 + 4 + 16 = 21.$$